

Manuscript outline

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Introduction

1. The most frequently observed biological impact of climate change is major changes in phenology
 - (a) Shifts in spring and autumn phenology modifies when seasons starts and ends
 - (b) Understanding the consequences of these shifts requires understanding how much and why growing seasons have changed
 - (c) Spring phenology trends and drivers
 - (d) Autumn phenology trends and drivers; shifts are weaker than springs'
2. Shifts in spring and autumn phenology supports the assumption that longer growing seasons increase tree growth
 - (a) Recent research shows that this may not be the case
 - (b) This has possible consequences on forest carbon cycle projections
 - (c) The different effects that shifting spring and fall events have on growth could impact these projections
3.
 - (a) Understanding these finds requires answering why trees do not grow more despite longer growing seasons.
 - (b) We don't understand how carbon allocation to above-ground biomass, despite it being one of the largest carbon sink
 - (c) Trees may be source or sink limited, which emphasizes the importance of understanding the temperature sensitivity relationship between growth and photosynthesis
 - (d) Growth may be more sensitive to varying environmental conditions than photosynthesis, which could mislead models.
 - (e) Hard nature of climate change makes understanding the internal physiological constraints and external constraints to growth critical
4. Growth drivers differences across species need to be considered
 - (a) Phenology and the relationship between growth and growing season length varies heavily accross species
 - (b) Polling all species together in carbon sequestrations models is a weakness
 - (c) We propose to use ground-based observations to address this issue
 - (d) These are useful to for the growth temporality of trees not suitable for experimental trials
 - (e) Cambial and leaf phenology are closely synchronized, so we can use them to as a proxy for start and end of season
5. How local adaptation could inform growth responses of species under future climate change
6. Understanding the constraints to growth requires defining the growing season and growth

- (a) Different definitions of GS: we propose to use calendar (number of days) and thermal season (heat accumulation)
 - (b) Xylogenesis throughout the growing season
 - (c) Tree rings as a proxy for growth
7. Question(s?): Do trees grow more with longer growing seasons?
- (a)
8. Objectives
- (a) Determine if trees grow more with longer thermal vs calendar season (ref: conceptual figure)
 - (b) Understand the calendar growing season relationship with growth and how it relates to the SOS and EOS
 - (c) Determine how growth changes across species and their provenances

Methods

1. Common garden set up and location
 - (a) Seed collection of 17 woody species across a latitudinal gradient
 - (b) Four sites Harvard Forest, the White mountains, Second College Grant of Dartmouth, College, and St. Hippolyte.
 - (c) Seeds germinated following standard germination protocols in 2016
 - (d) Seedlings planted at Weld Hill
 - (e) Design was a randomized 16 plot blocks, with shade cloth
 - (f) Plots weeded and watered throughout the study, pruned in the fall of 2020
2. Leaf phenology observations
 - (a) We observed leaf phenology twice per week (2018-2019) or once per week (2020)
 - (b) We used BBCH scale
3. Cookie collection
 - (a)

Results

Separate document

Discussion